

ROPE HAS A SPECTRAL BUDGET

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ABSTRACT

Structured reallocation of RoPE’s finite rotary budget produces large long-context gains. On a frozen billion-scale OLMo checkpoint, a model-relative derived allocation, separate from our training-time EVQ-COSH construction, raises nine-task 16K RULER from 0.56% to 60.47% (official Transformers YaRN: 7.94% on the same rows); a coarse task-label-free allocation reproduces the recovery, with no training or parameter updates. Under continued 750M training, EVQ-COSH lowers 16K PPL from 45.1 to 24.4 and raises strict 8K autoregressive exact retrieval from 0% to 77.5%; matched 1.485B and 8B adaptations further yield task-family length transfer and causal remote-source use. Writing $x_k = -\log \omega_k = a + Rz_k$ separates sampled support (a, R) from interior allocation z . A paired three-seed fixed-support intervention isolates the allocation effect; when a target-aware support policy may use L_{target} , the FMRoPE control reverses the tested ordering, identifying support and allocation as distinct but interacting levers. Exact phase-invariant sin/cos geometry explains how a fixed nominal budget can collapse onto a much smaller positional basis, and a convex surrogate yields closed-form zero-learned-parameter EVQ-COSH, unique within that surrogate. Across fully frozen, matched-adaptation, and from-training routes, structured allocation turns a finite spectral budget into usable context.

1 INTRODUCTION

A finite RoPE table has a spectral budget: its exactly $K = d_{\text{rot}}/2$ rotary pairs must allocate phase resolution across scales. On a frozen billion-scale OLMo checkpoint, a model-relative derived allocation raises nine-task 16K RULER from 0.56% to 60.47%, while official Transformers YaRN scores 7.94% on the same rows; a coarse task-label-free allocation reproduces the recovery without training or parameter updates. These model-relative frozen tables are separate from the EVQ-COSH construction used in training and adaptation.

Most RoPE optimisation is instead described through a scalar base, a spectral range, or a transform of the phases realised at longer positions. That framing leaves a separate finite-table choice implicit: where the fixed pairs place phase resolution inside their sampled range. We separate the choices by writing

$$x_k = -\log \omega_k = a + Rz_k, \quad z_0 = 0, \quad z_{K-1} = 1, \quad (1)$$

where k indexes the fixed K rotary pairs, a sets support location, R its span, and z the normalised interior allocation; geometric RoPE fixes $z_k = k/(K-1)$. Thus support (a, R) and interior allocation z are distinct coordinates: z decides which phase scales the fixed samples occupy and where resolution is placed.

We study structured allocation through three distinct intervention routes. A fully frozen route installs model-relative derived or coarse allocations with no parameter updates. Matched low-rank adaptation tests whether pretrained representations can exploit an EVQ-COSH basis, yielding task-family length transfer at 1.485B and causal remote-source use at 8B. From training, weights co-adapt to the installed table: in a 750M continuation, EVQ-COSH lowers 16K PPL from 45.1 to 24.4 and raises strict 8K autoregressive exact retrieval from 0% to 77.5%. These routes share a design principle; their estimands remain separate.

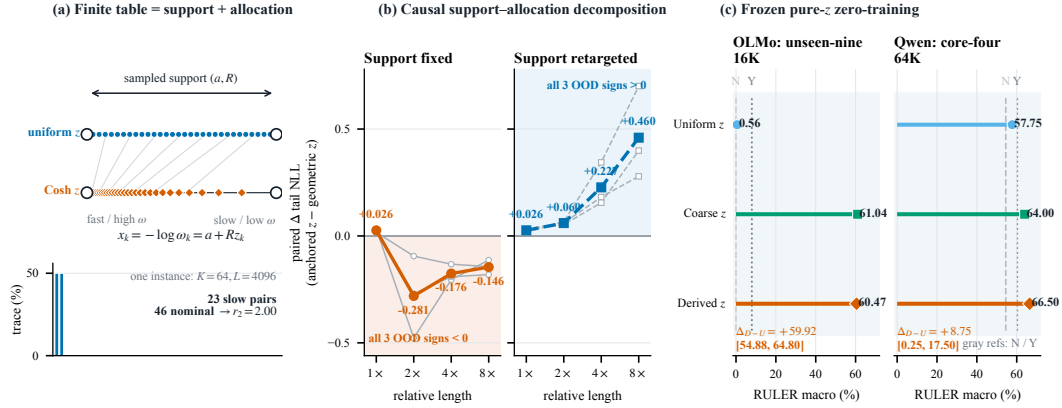


Figure 1: **A finite spectral budget is both identifiable and actionable.** (a) Support (a, R) does not determine the interior allocation z ; the lower inset gives one representative standard-RoPE slow block, where 23 pairs (46 nominal dimensions) have block-whitened $r_2=2.00$ under the stated uniform separation prior. (b) Across three paired training seeds, anchored EVQ-COSH minus paper-faithful FMROPE tail NLL is negative at every OOD length when support is fixed and only the 30 interiors move; allowing the declared L_{target} to retarget support reverses every OOD sign. The two conditions separate allocation from support rather than stage a method race. (c) On frozen OLMo and Qwen checkpoints, the coloured block holds support, amplitude, checkpoint, rows, decoder, precision, and hardware fixed: changing only z produces the large zero-training RULER gains. Grey Native (N) and official Transformers YaRN (Y) rows retain their own support or transport identities.

The causal foundation separates allocation from support selection. Across three paired 151.9M seeds, anchored EVQ-COSH and paper-faithful FMROPE share sampled extrema and log-span while only the 30 interiors move; every OOD length favours the reallocation in every seed. In a separate target-aware condition, the declared L_{target} may retarget support and the ordering reverses. Together the two conditions identify support and structured allocation as distinct but interacting controls of the same finite table (§2.1).

This distinction matters because geometric spacing is highly redundant at the slow end. For $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions yet have block-whitened Rényi-2 effective rank 2.00. These channels can still carry content, but as positional functions they converge toward the same two-dimensional subspace. A geometric table therefore spends more than one third of its rotary pairs on near-duplicate slow positional directions. This is the spectral-budget problem studied here.

Optimising a convex surrogate over z yields *Exact Variational Quantization with a Cosh density* (EVQ-COSH), a closed-form table whose inverse-CDF quantiles require no learned positional parameters. The solution is unique within that surrogate.

Figure 1 defines the design object, separates the support and allocation interventions, and shows the mature zero-training consequence. The experiments then follow the practical order above—fully frozen, matched adaptation, and from training—across full and scarce rotary budgets and a bidirectional 3D video DiT (§4).

Contributions. (i) We demonstrate structured allocation across three model-building stages, led by a zero-training mature-checkpoint intervention and supported by strong long-context likelihood, downstream, retrieval, and causal source-use results. (ii) We separate support (a, R) from interior allocation z : a paired three-seed fixed-support intervention identifies allocation, while target-aware support retargeting reverses the tested ordering and establishes their interaction. (iii) We give an exact phase-invariant account of the full sin/cos basis: a spectral-budget identity, a slow-frequency collapse law, the weight-table co-adaptation boundary, and a frozen-transplant obstruction. (iv) We solve a convex allocation surrogate in closed form to obtain EVQ-COSH with no learned positional parameters, and test that construction across training and matched-adaptation regimes while keeping the separate frozen route explicit.

2 SUPPORT-ALLOCATION CAUSAL DECOMPOSITION

2.1 FIXED SUPPORT ISOLATES STRUCTURED ALLOCATION

A clean interior-allocation intervention. Within each of three 151.9M training seeds, both arms share architecture, initialisation, token order, optimiser, LR schedule, global batch, a 499,974,144-token budget, and 32 evaluation anchors. The baseline is FMROPE at fixed training support, implementing Oka et al. (2026b, §6.1) with $\theta=L_{\text{train}}=256$ and geometric exponents. Anchored EVQ-COSH ($\tau=4$) normalises its quantiles to the same extrema and log-span, so only the $K-2=30$ interior frequencies move while retaining the deployed Cosh table’s z . Every training seed favours anchored EVQ-COSH at every OOD length. Anchored EVQ-COSH minus FMROPE NLL is $+0.026$ at the 256-token training length and $-0.281/-0.176/-0.146$ at 512/1K/2K (Fig. 1c). No geometric base can reproduce this intervention once the sampled support is pinned: *fixed support + different $z \Rightarrow$ different trained-model behaviour*. Thus z is a separately identifiable coordinate, not a disguised base change.

Target-aware support identifies the second coordinate. The second condition permits the declared L_{target} to retarget each arm’s sampled support while preserving its normalised interior shape, giving the paper-faithful FMROPE control its intended target-aware input. The ordering then reverses in all three seeds: anchored EVQ-COSH minus FMROPE NLL is $+0.060/+0.227/+0.460$ at 512/1K/2K. The two conditions answer different causal questions. Fixed support isolates the effect of structured interior allocation; target-aware retargeting shows how support selection changes which allocation is effective. Their reversal establishes support and allocation as distinct but interacting coordinates of the same finite table. The full protocol, per-seed contrasts, and both support conditions are in App. C.

Configuration and shape breadth. The fixed-support result is not confined to one base, head width, or Cosh strength. A pre-specified 50.9M factorial crosses two bases, two training lengths, three head dimensions, and three paired seeds while pinning sampled endpoints and log-span in every arm. Three Cosh strengths and a deformation-matched exponential provide four structured non-geometric curves. The strongest Cosh arm improves the weighted OOD objective in 10/12 configurations; the matched exponential does so in 9/12 and closely tracks the formula-Cosh arm. This broadens the finding from one curve to a family of structured allocations: Cosh supplies the analytic construction, while the 151.9M intervention above remains the effect-size owner. The complete matrix is in App. Table 11; model-scale persistence is tested separately in §4.4.

3 THEORY

Finite RoPE design is a constrained basis-allocation problem. With pair count and sampled support fixed, a table chooses where to place phase resolution. We first account for that finite budget exactly, then locate the slow-end redundancy of geometric sampling and derive EVQ-COSH as a closed-form reallocation. Finally, we connect the designed basis to the stage at which weights exploit it; capability is established by the controlled interventions, not inferred from static geometry.

3.1 A FINITE SPECTRAL BUDGET IS A DESIGN PROBLEM

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$, let d_{rot} be the number of rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs ($d_{\text{rot}}=d_{\text{head}}$ for standard MHA, smaller for compressed-RoPE architectures such as MLA).

We reserve z for the support-normalised coordinate in Eq. (1). Normalising a sampled geometric table gives $z_k = k/(K-1)$, so fixing (a, R) fixes the geometric family and a non-uniform interior allocation is a distinct fixed-support design. Standard implementations use raw exponents $u_k = k/K$, deployed EVQ-COSH uses midpoint quantiles $u_k = (k+1/2)/K$, and endpoint-inclusive diagnostic grids are labelled explicitly.

A rotary pair contributes $f_{\omega}(\Delta) = C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to the relative-position attention logit, with C, D set by content. The object attached to ω is therefore the two-dimensional subspace

$$V_{\omega} = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}, \quad (2)$$

not a cosine feature: an analysis that fixes one content phase sees a slice of V_ω and is not invariant to rotation inside the pair. Two channels are redundant to the extent that their subspaces are. With $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, Gram matrices $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ over a separation prior, and the whitened cross-Gram $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ whenever $S_\omega, S_\nu \succ 0$, the singular values of $Q_{\omega\nu}$ are the canonical correlations between V_ω and V_ν , and

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1] \quad (3)$$

is invariant to phase rotation inside a pair and to any change of basis of either subspace. For $\Delta \sim \text{Unif}[0, L]$, $H_{\omega\nu}$ has a closed form in $\sin t/t$ and $(1 - \cos t)/t$ at $t = (\omega \mp \nu)L$ (App. A).

Block-whitening every pair gives a global correlation Gram Γ with diagonal blocks I_2 . Write $\bar{c} = [K(K-1)]^{-1} \sum_{i \neq j} c_{ij}$.

Theorem 1 (Spectral budget identity). $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$, so the Rényi-2 effective rank of the positional basis is exactly

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (4)$$

Practically, this identity exposes the design budget. For fixed K , sampled support (a, R) , and separation measure μ , substituting $\omega_k(z) = \exp[-(a + Rz_k)]$ into the block-whitened Gram $\Gamma_\mu(z)$ changes how the same trace $2K$ is distributed across positional directions. Block whitening removes per-channel energy and conditioning: the theorem characterises the basis’s Rényi-2 effective dimension, while trained-model protocols establish what that basis enables.

3.2 SLOW-END REDUNDANCY GIVES A DESIGN DIRECTION

Proposition 2 (Low-frequency collapse). As $\omega L \rightarrow 0$, $V_\omega \rightarrow \text{span}\{1, \Delta\}$. For slow channels $x=\omega L$, $y=\nu L$, and $\epsilon = \max\{|x|, |y|\} \rightarrow 0$, $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$: slow subspaces approach a common two-dimensional limit at fourth order. Under the softmax metric for a fixed p with nondegenerate support on at least three distances, $F = \text{diag}(p) - pp^\top$ annihilates the constant direction, and the centred limit is $\text{span}\{\Delta - \mathbb{E}_p \Delta, \Delta^2 - \mathbb{E}_p \Delta^2\}$.

Practically, geometric slow-end sampling provides diminishing new positional directions. This motivates thinning that region while preserving support and multiscale coverage; the channels may still carry content, and the static rank is not a behavioural selector. As a representative severe-regime instance, at $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 standard-grid pairs with $\omega L \leq 1$ occupy 46 nominal dimensions but have block-whitened $r_2=2.00$ under the uniform separation prior. The full-table identity also applies outside this asymptotic regime; additional views appear in App. Fig. 4.

3.3 A CLOSED-FORM REALLOCATION

A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a finite allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$; geometric RoPE is $\rho \equiv 1$. The exact canonical-correlation objective is oscillatory, so we use the convex modelling surrogate

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (5)$$

with $\alpha > 0, \beta \geq 0$. The first term limits channel-load concentration; the Green kernel $\min(\phi, \psi)$ penalises pairs that are both slow, following the direction of Proposition 2.

Theorem 3 (Closed-form optimum under the surrogate). Under (5) with $\alpha > 0, \beta \geq 0$ and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}; \quad (6)$$

at $\beta=0$ it is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$.

Practically, the theorem turns the slow-end design direction into an analytic allocation rather than a search problem. The following inverse CDF instantiates that continuous solution as a finite table.

Corollary 4 (Closed-form quantiles). $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}, \quad (7)$$

which tends to u_k as $\tau \rightarrow 0$, recovering the uniform-log midpoint grid.

For $\tau > 0$, ρ_τ decreases across log-frequency and crosses the uniform density once (Lemma 7); the quantiles therefore place more samples on the fast side and thin the redundant slow side. This gives EVQ-COSH explicitly, without learned positional parameters or numerical search. Its uniqueness is conditional on (5); the operating rule, finite multiplier evidence, grid conventions, and exact-range endpoint embedding are in Apps. A.9 and C. The cosine-feature diagnostic is reported separately in App. A.6; trained interventions establish behaviour.

3.4 A DESIGNED BASIS MUST BE EVALUATED AT ITS INSTALLATION STAGE

A designed table fixes the positional basis; model capability depends on the coefficients that exploit it. Controlled fixed-support protocols show that designed reallocations can strongly improve capability in the tested protocols. The training intervention in §2.1 evaluates anchored EVQ-COSH, while the fully frozen study in §4.1 evaluates separate derived and coarse profiles; they supply likelihood and capability evidence, respectively.

The installation stage is visible in a frozen 50M weights-by-table crossing (Table 1). The self-consistent Geo and EVQ-COSH systems have PPL 7.14 and 7.16, while both mismatched cells degrade. In particular, installing EVQ-COSH under Geo-trained weights raises block-whitened r_2 from 4.57 to 12.54 but worsens PPL from 7.14 to 76.20; the loss interaction is -3.537 (bootstrap CI $[-5.17, -3.04]$).

Table 1: **Weights co-adapt to their training-time frequency table.** Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head–query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

Thus static geometry diagnoses the designed basis but does not rank trained systems: weights learn coefficients in the basis installed during training or adaptation.

Theorem 5 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta) B = \mathcal{R}_\Omega(\Delta)$ for all Δ in an open interval containing zero, then Ω' and Ω have the same frequency multiset, up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Practically, a nontrivial new table cannot be made exactly function-equivalent by fixed, position-independent invertible Q/K maps: the design must be evaluated with the weights, adapter, or operator that will use it. At $\Delta=0$ the hypothesis gives $A^\top B = I$; differentiating the resulting similarity at zero preserves the generator spectrum $\{\pm i\omega_k\}$ (App. A). The theorem does not exclude approximate finite-data adaptation or broader paths, which are tested in §4.3.

Together, the budget identity defines what can be reallocated, the collapse law locates redundant slow-end sampling, and the variational theorem supplies a closed-form table before its coefficients are learned. The fixed-support and lifecycle experiments test designed allocations with weights fully frozen, under matched adaptation, and through from-training co-adaptation (§4).

4 THE ALLOCATION COORDINATE ACROSS MODEL-BUILDING REGIMES

We intervene on the allocation rule at three stages of the model lifecycle. With a mature checkpoint fully frozen, only the installed allocation changes; under matched low-rank adaptation and from training, the resulting weight changes are responses to that allocation rather than separate method variants. We keep likelihood, capability, and uncertainty units separate throughout.

4.1 FROZEN MATURE MODELS: PURE- z ZERO-TRAINING INTERVENTION

Allocation alone changes a mature model. At matched support, amplitude, checkpoint, rows, decoder, precision, and hardware, changing only interior z raises OLMo nine-task RULER at 16K from 0.56% to 60.47%; a coarse label-free allocation reaches 61.04%. On Qwen at 64K, the corresponding fixed-support change raises core-four RULER from 57.75% to 66.50% (Fig. 2). The intervals for derived minus uniform allocation are [54.88, 64.80] and [0.25, 17.50] points, respectively. These uniform, coarse, and derived mature tables are three values of the same fixed-support coordinate; they are not EVQ-COSH tables, and the coarse-versus-derived comparison does not identify a unique profile (App. E.2).

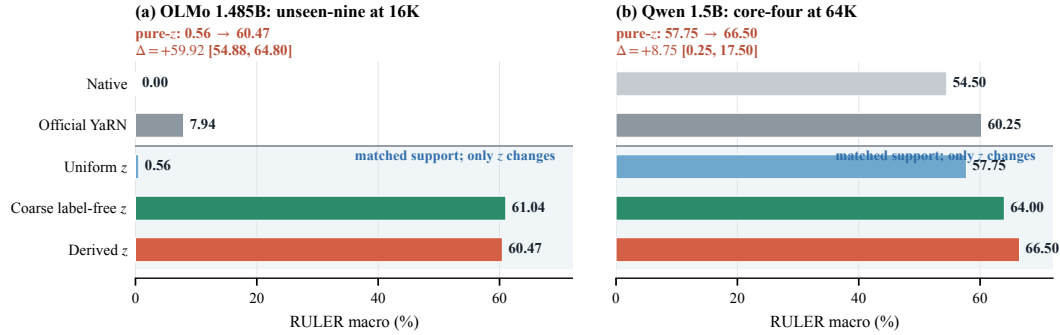


Figure 2: **Changing only interior allocation can transform a frozen checkpoint.** Grey reference rows retain their own support or transport; the shaded block holds support, amplitude, checkpoint, rows, decoder, precision, and hardware fixed while changing only z . Derived minus uniform allocation is +59.92 points ([54.88, 64.80]) on OLMo and +8.75 points ([0.25, 17.50]) on Qwen. Each task has 20 rows; intervals condition on each fixed checkpoint and task set through evaluation-row resampling. Exact values and the coarse-control contrasts are in App. E.2.

For implementation, given Native frequencies ω_k , extension factor s , and fixed displacement weights $m_k \in [0, 1]$, the mature intervention installs

$$\hat{\omega}_k = \omega_k [(1 - m_k) + m_k/s], \quad \hat{z}_k = \frac{-\log \hat{\omega}_k - \hat{a}}{\hat{R}}, \quad (8)$$

with the fast and slow endpoints pinned. Equation (8) defines the pure- z block above; the complete deployment policy below is a separate bundled intervention.

4.2 ZERO-TRAINING DOWNSTREAM CAPABILITY AND NATIVE-WINDOW RETENTION

Building on the pure- z result, the zero-training system bundles one frozen factor-four long table, a fixed long attention amplitude, and a session-static Native/long route chosen before prefill. It adds no learned parameters or training tokens, but its combined gain must not be assigned to allocation alone. Table 2 separates its likelihood, capability, downstream, and routing endpoints; none of these metrics is pooled.

The route retains the exact Native path when a request fits the model window; otherwise it uses one deployment-frozen factor-four profile for the session. This practical policy is neither the pure- z contrast in Fig. 2 nor the EVQ-COSH construction used in the adapted and from-training studies. Full task-level results, row identities, and the allocation-versus-routing controls are in App. E.2.

Table 2: **Zero-training system endpoints.** The bundled policy chooses one Native or frozen long route before prefill and retains it for the session. Rows use different metrics and experimental units and are not pooled. The policy is distinct from the pure- z block in Fig. 2 and from EVQ-COSH.

Endpoint	Reference	Bundled policy	Evidence role
FineWeb-Edu final-1,024 tail NLL, 4K ($n=128$)	Native 2.7538; static long 2.8774	2.7538 (Native route)	in-window routing
PG-19 tail NLL, 8/16K	official YaRN 3.4404/3.7955	3.1060/3.0977	natural likelihood
RULER-13 macro, 8/16K	official YaRN 23.82/5.88%	67.72/54.40%	task capability
Qasper (Dasigi et al., 2021) / 2Wiki token-F1, full 200	official YaRN 18.03/25.69%	24.57/26.66%	natural downstream

4.3 MATCHED ADAPTATION: LENGTH TRANSFER AND CAUSAL SOURCE USE

Matched adaptation asks whether pretrained representations can learn to use an EVQ-COSH allocation. Starting from protocol-matched Native and EVQ-COSH Stage-A parents, the 1.485B selective-Q/K arms freeze inherited V/O LoRA and continue only Q/K for 300 steps with identical data, order, optimiser, trainable count, and phase exposure. On 200 held-out 2WikiMultiHopQA rows per length, Native/EVQ-COSH token-F1 is 0.07/21.48% and 0/8.57% at 8/16K, versus 25.99/24.84% at 4K (Ho et al., 2020; Bai et al., 2024). An independent 13-family RULER continuation gives the same task-family crossover: 2.02/31.63% and 0.38/5.03% at 8/16K, versus 72.19/42.44% at 4K (Hsieh et al., 2024). Training and evaluation rows are disjoint within each task family (App. E).

At 8B, a single matched rank-64 Q/K/V/O LoRA pair provides a complementary causal endpoint. At true 16K, the EVQ-COSH arm raises median target-block hit@16 from 18.75% to 64.06%; position-preserving removal of the remote gold block changes answer NLL by -0.0095 / $+1.5055$ (Fig. 3). Thus the adapted EVQ-COSH model uses the remote source in these cases; this routing/probability result is distinct from autoregressive task accuracy.

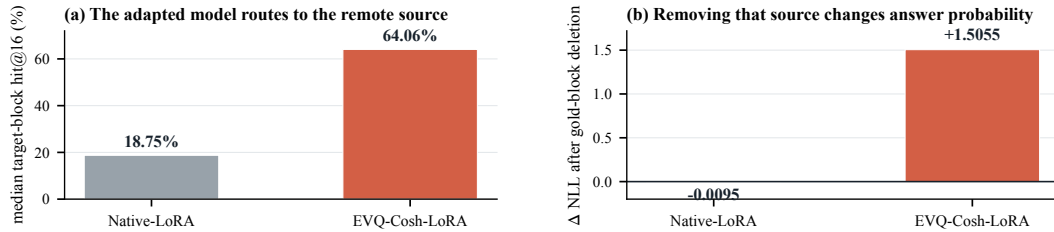


Figure 3: **Matched adaptation converts the new basis into causal remote-source use.** In one matched trained pair, ten frozen true-16K cases measure (a) median target-block hit@16 over 32 retrieval heads and (b) the answer-NLL change after deleting decode attention to the remote gold block from every head while preserving prompt tokens, rotary KV indices, and block geometry. These are routing and teacher-forced probability endpoints, not training replications or autoregressive accuracy (App. E.3).

4.4 FROM TRAINING: CO-ADAPTATION, SYSTEMS, AND IN-WINDOW BEHAVIOUR

The paired three-seed intervention in §2.1 owns the fixed-support causal claim. The studies in Table 3 instead test whether an installed allocation rule remains consequential under scarce rotary channels, continued or from-initialisation full-parameter training, and bidirectional video. The fixed-support studies own causal attribution; these rows test stage-specific persistence, with each protocol’s execution boundary retained in the table and appendix.

The text protocols exhibit a recurring in-window/long-range crossover, while the video-DiT comparison improves both training and extrapolated frames. These rows therefore establish architecture-,

Table 3: **From-training and system breadth.** Geo/EVQ-COSH entries are reported within each protocol; Δ NLL is EVQ-COSH minus Geo. Rows have different experimental units and endpoints and are not pooled. The allocation rule is the intended scientific intervention; causal attribution is owned by the fixed-support studies, while these rows establish stage-specific persistence.

Protocol (unit)	Original window	Longer positions / capability	Maximum role
432M MLA, PPL $35.4/35.8$ at PPL $138.8/95.6$ at 16K			scarce-channel system
$K=16$ (3 seeds) 8K			persistence / capability
750M continuation PPL $22.0/22.3$ at PPL $45.1/24.4$ at 16K; AR exact 0/77.5% (1 pair, seed 42) 4K at 8K			pretraining-scale persistence
1.485B from initial- Δ NLL $+0.0724/+$ Δ NLL $-0.0437/-$ 0.1351 at 8/16K isation (1 same-init 0.0381 at 2/4K comparison)			cross-modal breadth
129.6M video DiT train-frame (seed 42) -21%	MSE all/far extrapolated MSE $-16\%/-$ 35%		

scale-, and modality-specific persistence, not a pooled effect or scaling law. Complete PPL trajectories, strict generation endpoints, and per-seed MLA values are in Apps. B, E, and F.1.

5 RELATED WORK

Positional encoding can determine whether a transformer generalises beyond its training length (Kazemnejad et al., 2023). RoPE extensions rescale positions, change a scalar base, transport frequency bands, or remap remote phases (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025; Tian et al., 2026; Tang et al., 2026). FMRoPE provides a target-aware support/base construction for a chosen length (Oka et al., 2026b). Its paper-faithful arm serves here as the geometric causal control, not a competing method: pinning its endpoints and log-span lets us change only the interiors. Frequency-dependent transport can likewise realise a non-uniform z while moving support or phase. Base analyses bound the geometric family’s attainable context (Xu et al., 2024; Liu et al., 2024), while learned use of a supplied grid adds a data-dependent view (Wu et al., 2026).

Other work measures unequal use of RoPE bands, removes or clips selected rotations, learns input-dependent angles or frequency tables, or replaces the fixed table with a larger positional function (Barbero et al., 2025; Oka et al., 2026a; Chiang & Yogatama, 2025; Chen et al., 2025; Li et al., 2026a; Wertheimer et al., 2026; Zhang et al., 2026; Movahedi et al., 2026; Wang et al., 2024; Zheng et al., 2024; Li et al., 2024; Hua et al., 2025; Li et al., 2026b). LeRoPE and AdaRoPE make allocation explicit through learned non-geometric frequencies (Karypis et al., 2026; Wang et al., 2026); LeRoPE’s in-window improvements provide convergent evidence that allocation matters beyond extrapolation. Its learned intervention complements our matched fixed-support controls, which retain all K canonical rotary planes, identify z causally, and derive a closed-form table without frequency search or learned positional parameters.

6 DISCUSSION

Designed interior allocations produce strong capability changes in fully frozen checkpoints, remain actionable under matched low-rank adaptation, and alter outcomes through training-time weight-table co-adaptation. These three intervention stages establish z as a controllable RoPE design coordinate. Exact full-subspace geometry explains how allocations change the available positional basis; the weights-by-table crossing and transplant obstruction explain why the intervention stage matters.

Support and allocation are complementary, interacting coordinates. The fixed-support controls hold K , both endpoints, and the RoPE operator constant to identify z alone, while target-aware support retargeting reverses the tested ordering. A realised table therefore reflects both sampled support (a, R) and the interior allocation z of its fixed K pairs; frequency transport may change both at once.

Across the autoregressive text protocols that evaluate both the original training or Native window and longer positions, the comparisons reveal a within-protocol crossover: the baseline is stronger

near the original window, while the reallocated table overtakes it beyond that window. This recurring behavioural signature of finite-budget reallocation directly motivates the Native/long session route. The video DiT shows another regime: both training-frame and extrapolated-frame MSE improve (-21% on training frames and -35% on far extrapolated frames), and its pre-specified diagnostic connects the gain to in-window phase coverage. Together with LeRoPE’s learned in-window allocations, these results make a single zero-training allocation that itself improves Native-window and long-range capability—rather than preserving the former through routing—the natural next method direction.

The practical consequence is that a finite RoPE table should be specified by both its sampled support and its interior allocation. A shared base, endpoint, or pair count does not by itself define a matched finite table. Treating z as an explicit design object makes allocation measurable and controllable across training, adaptation, and deployment.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

Equation (7) specifies the default midpoint EVQ-COSH table as a pure function of (τ, K) , with no learned parameters or auxiliary loss. Anchored support embeddings, endpoint $u_k=k/K$ quantisation in the mature OLMo EVQ-COSH protocols, and the inclusive static diagnostic are labelled where used; each retains the same continuous Cosh allocation law but defines a different finite grid or support. Equation (37) is the full-RoPE EVQ-COSH reference rule; every EVQ-COSH protocol reports its realised τ explicitly, including the separately fixed MLA and video settings. The mature zero-training uniform/coarse/derived allocation tables instead follow Eq. (8) and are specified in App. E. Complete proofs of Theorems 1, 5 and 3, Proposition 2, Corollary 4 and Lemma 7, together with every stated assumption, are in App. A. The identification protocol, including how sampled extrema and log-span are pinned, the exact pre-specified factorial design, and the corpora used by each arm, is in App. C; the mature-scale adaptation protocols, module budgets, phase-exposure mixtures, and evaluation harness settings are in App. E; remaining architectures and hyperparameters are in App. B. Seed scope is stated with every result. The anonymous supplement includes the exact-range training entripoint and three-seed aggregate, the frozen figure generators, curated mature-protocol snapshots, and CPU tests, with a reviewer-facing file map and commands.

AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

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A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_*(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_*(s) - b_*(d) \\ b_*(s) + b_*(d) & a(d) - a(s) \end{bmatrix}, \quad (9)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. The metric is consequently attached to the positional subspaces themselves, not to a chosen sine/cosine phase convention.

Proof of Theorem 1. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (4). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 2. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (10)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (11)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 , rather than comparing the unwhitened Taylor coefficients. Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (12)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (13)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (14)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1} M) + O(\epsilon^6) = \frac{19}{12600} (x^2 - y^2)^2 + O(\epsilon^6), \quad (15)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \longrightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{2\overline{\cos(\omega\Delta)}}{\omega^2} \longrightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (16)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. This is a limit for a declared fixed measure; it does not identify which slow channels a trained model uses. $\square$

The reported $r_2=2.00$ for the 23 slow pairs uses the standard $u_k=k/K$ grid and the uniform separation prior above. Other priors change that static numerical Gram. The fixed-range trained-model contrast of §2.1 instead follows its paired empirical protocol.

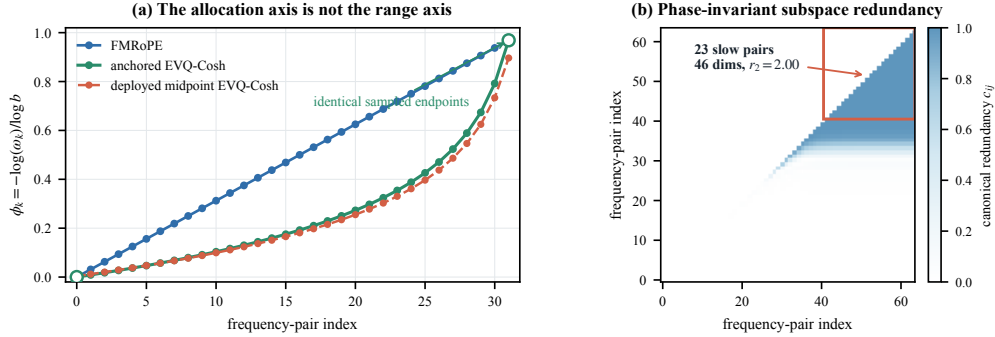


Figure 4: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

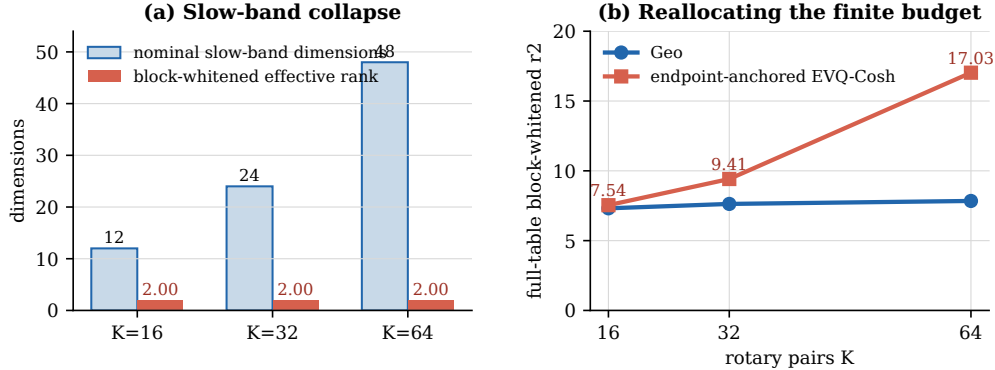


Figure 5: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows. These are static positional-basis diagnostics; the trained effect is identified separately in App. C.

A.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an

exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

Proposition 6 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (17)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (18)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (19)$$

so every entry of the cross-Gram in (9) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_*((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (18) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k\Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 6 attains the algebraic rank maximum on the declared interval while imposing an exact recurrence just outside it. It does not say that K_{par} is a global capacity over arbitrary frequencies, nor does it survive a change of measure: the causal separation histogram is triangular rather than uniform and does not make the same lattice orthogonal. The exact parity construction and the small-model checks are reproduced by `analysis/full_rope_audit/verify_small_models.py`; the uniform-versus-causal measure comparison is exposed by `scripts/analysis/third_axis_ceiling.py-decompose`. These are static basis statements, not LM-quality or extrapolation predictions.

A.3 TWO STATIC ORDERING COUNTEREXAMPLES

Even within a fixed declared measure, a cheaper static proxy need not order tables like the full basis, and an ordering at one interval need not persist at another. The following two finite tables are existence counterexamples; no claim of continuous global optimisation is needed.

Cosine-only collision can choose the lower-rank table. Use $L=4096$, $b=500,000$, $K=64$, continuous uniform $\Delta \in [0, L]$, and pin both physical endpoints to $[2 \times 10^{-6}, 1]$. Let A be the best-found schedule returned by the declared heuristic for cosine-only collision and B the corresponding best-found schedule for full-subspace collision, using their distinct harmonic-augmented candidate pools. The cosine proxy prefers A ,

$$C_{\text{cos}}(A) = 2.62 \times 10^{-9} < 1.44 \times 10^{-8} = C_{\text{cos}}(B), \quad (20)$$

but the phase-invariant block-whitened basis has the opposite order,

$$r_2(A) = 64.48 < 128.00 = r_2(B). \quad (21)$$

Here C_{cos} is the mean squared normalised cosine-cosine overlap. The counterexample isolates why one phase slice cannot replace the full two-plane object: a small cosine overlap can hide redundancy in the sine and cross terms.

Full-subspace collision can reverse across lengths. Under the same base, endpoints, and training-interval convention, the tracked counterexample search emits two fixed $K=16$ tables A and B . Writing C_M for their mean phase-invariant full-subspace collision when the uniform measure spans $[0, M]$, their ordering is

$$C_L(A) = 0.45837 < 0.61695 = C_L(B), \quad (22)$$

$$C_{2L}(A) = 0.45834 > 0.41132 = C_{2L}(B), \quad (23)$$

$$C_{4L}(A) = 0.45830 > 0.24763 = C_{4L}(B). \quad (24)$$

Recomputing the same schedules on discrete uniform separations $\Delta = 0, \dots, M-1$ preserves the reversal. Thus even the correct phase-invariant collision is conditional on the declared interval; minimising it at L does not supply an extrapolation ordering at $2L$ or $4L$.

Numerical owner and reproduction convention. The digits in Eqs. (21)–(24) are frozen outputs produced by `scripts/analysis/full_rope_collision_audit.py` with `-length 4096 -base 500000 -candidate-grid 1025 -random-schedules 3000`; the counterexample RNG seed is 20260819 and the JSON stores each schedule and its float64 hash. These are best-found points returned by the declared greedy-plus-exchange heuristic in metric-specific candidate pools, not certified pool or continuous optima. The first inequality is also reported with raw full entropy rank $93.44 < 120.03$; the stable-rank ordering above is the quantity governed by Theorem 1.

Together, the parity recurrence and the ordering reversals separate three questions: dimension over a declared interval, multiscale coverage outside it, and behaviour after training. The fixed-support experiments evaluate the last question, while Table 1 shows why learned coefficients must stay with the basis that trained them.

A.4 POST-HOC TRANSPLANT OBSTRUCTION

Proof of Theorem 5. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace; it need not preserve the original two-dimensional block decomposition. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

This exact statement covers *fixed, position-independent, invertible* compensation. Approximate re-training on finite data, nonlinear maps, and changes to V/O or the residual path are separate intervention classes.

A.5 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 3 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (25)$$

the main-text collision functional is (5), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (26)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e.}, \int_0^1 \rho = 1\}$. The $\alpha\|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (27)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (28)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (29)$$

since differentiating $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha\rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha\rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (28) and (29) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (30)$$

which is main-text Theorem 3. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1) = \tau / \sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler-Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (30).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (31)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

Why this branch. The homogeneous Cosh solution has an analytically invertible CDF, turning the continuous optimum into an explicit finite table. The functional test below checks the resulting allocation against a closed-form cosine-feature slice, and the trained comparisons supply the behavioural test.

A.6 COSINE-SLICE DIAGNOSTIC FOR THE SURROGATE-DERIVED TABLE

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ supplies the closed-form density. We validate its allocation functionally against the closed-form cosine-feature kernel $K_{\text{cos}}(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This tests whether the same direction holds for the declared cosine-feature slice.

Table 4 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute K_{cos} at the discrete allocation points for Geo and EVQ-COSH, then evaluate the normalized collision

score $C = \sum_{i < j} K_{ij}^2 / (K_{ii} K_{jj})$ and Shannon effective rank (the exponential of the cosine-feature Gram eigenvalue distribution). The inclusive grid $u_k = k / (K - 1)$ exposes the complete diagnostic interval. This slice-level static diagnostic is separate from the phase-invariant full-subspace geometry of Theorem 1; trained midpoint tables are evaluated in §4. The 12 reported configurations comprise a six-length text sweep at fixed $b=500K$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$.

Table 4: Cosine-slice surrogate check on the inclusive $u_k=k/(K-1)$ diagnostic grid. EVQ-COSH reduces collision and increases Shannon effective rank of the cosine-feature Gram under the closed-form kernel K_{cos} in all 12 tested configurations. Theorem 1 supplies the separate phase-invariant full-subspace geometry; §4 evaluates trained midpoint tables.

Configuration	Collision C		Shannon rank	
	Geo	EVQ-COSH (Δ_C)	Geo	EVQ-COSH
<i>Text: $K=32$, $b=500K$, sweep L</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32$, $L=2048$, sweep b</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16$, $L=32$, sweep b</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

EVQ-COSH reduces collision under K_{cos} in all 12 configurations (24%–92%), and Shannon effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.7 SINGLE-CROSSING BUDGET SHIFT

Lemma 7 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (32)$$

Proof of Lemma 7. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (32). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1 - \phi)) / \sinh \tau$ at $\phi_L = \log(L/2\pi) / \log b$ and $\phi_M = \log(M/2\pi) / \log b$. $\square$

A.8 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 8 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (33)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (34)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (35)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (33). $\square$

An immediate corollary follows.

Corollary 9 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (36)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity checks the two surrogate energy terms exactly; trained-loss evidence is supplied by §4.

A.9 ASSUMPTION-BOUND OPERATING CALCULATION AND FINITE TESTED RECEIPT

The variational theorem selects the Cosh *shape*; it does not select a finite strength for every model or protocol. The implementable reference used in the standard full-RoPE text experiments is

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (37)$$

Here $c=1$ is the pre-training zero-search convention, not a fitted optimum; each protocol records its realised τ , grid, support, and rotary width.

Assumption-bound calculation. The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45 d_{\text{head}}} + O(\tau^6), \quad (38)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (39)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1 - \phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (40)$$

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives only the leading balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (41)$$

Thus $d_{\text{head}}/\sqrt{L}$ is an assumption-bound scaling factor, while the coefficient remains utility-, base-, and convention-dependent. The reported experiments use the exact Cosh warp, including where small- τ truncation is not accurate; Eq. (41) is not a behavioural selector, a support-invariance statement, or a derivation of $c=1$.

Finite tested receipt. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at $0.75\times$, $1.25\times$, or $1.5\times$; the independent exact-range factorial tests $0.75\times$, $1.00\times$, and $1.25\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entryptpoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. These finite points support a fallible zero-search operating prior relative to the tested Geo baseline. They do not establish a continuous basin, basin bounds, interpolation guarantees, near-optimality, or a global optimum.

A.10 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (42)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (43)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (44)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (45)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoff, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (46)$$

This is the finite-channel mechanism: a shaped ρ can reduce the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Dependence of the bounds on channel count. The standard MHA configuration has $d_{\text{head}}=d_{\text{rope}}=64$ and $K=32$; the MLA study has $d_{\text{head}}=64$, $d_{\text{rope}}=32$, $d_{\text{nope}}=32$ and $K=16$. Halving K scales the K^{-2} component of this distortion bound by $4\times$ and the K^{-1} component by $2\times$. The MLA experiment (§F.1) supplies the corresponding trained-model test at $K=16$.

Large- τ regime. At large τ , $m_\tau=\tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B EXPERIMENTAL DETAILS

B.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model protocols are documented separately in Apps. C and E. This section records the scratch-model and cross-modal experiments that remain supporting evidence. The learned-frequency comparator appears in App. D.

B.2 REPRODUCIBILITY SNAPSHOT

The evidence ladder separates causal identification from architecture, scale, adaptation, frozen-checkpoint, and cross-modal consequences. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 5: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime and intervention	Unit, endpoint, and appendix
Allocation identification	151.9M scratch; fixed support, 30 interiors	3 training seeds; tail NLL; App. C.1
Co-adaptation replication	151.9M frozen crossing; weights $\times$ runtime table	2 training seeds; tail NLL; App. C.2
Shape/configuration breadth	50.9M factorial; fixed support, 5 allocations	12 configs $\times$ 3 seeds; weighted OOD NLL; App. C.3
Scarce rotary budget	432M MLA-style; Geo/EVQ-COSH, $K=16$	3 training seeds; PPL; App. F.1
Full-parameter scale	750M/1.485B; continued/from-init Geo/EVQ-COSH	one comparison each; PPL/retrieval; Apps. B.3/E.1
Frozen allocation	OLMo/Qwen; same support and amplitude	20 rows/task; RULER; App. E.2
Zero-training deployment	OLMo; frozen long table, gain, session route	fixed checkpoint/rows; NLL/RULER/F1; App. E.2
Adaptation/source use	1.485B/8B; matched LoRA/source block	one trained pair; QA/RULER/NLL; Apps. E.1/E.3
Cross-modal scope	129.6M video DiT; temporal Geo/EVQ-COSH	1 training seed; denoising MSE; App. B.5

Table 6 lists the training hyperparameters shared across the supporting text experiments.

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

Table 6: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 2–4 (750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Table 7: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

B.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses midpoint quantisation with $\tau=1.5$. Both arms are full-parameter continuations from the shared checkpoint.

B.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T} = 2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ ($0.53\times$), which selects the supporting video configuration.

B.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 8 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames, extending the scope check to bidirectional 3D RoPE.

Table 8: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

B.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 1 separately measures incompatibility after a frozen post-hoc swap.

C IDENTIFICATION PROTOCOL AND PROVENANCE

C.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMRoPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes

only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Concretely, let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R\frac{k}{K-1}\right). \quad (47)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed*: each checkpoint retains its training range. *target-matched*: the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 9: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

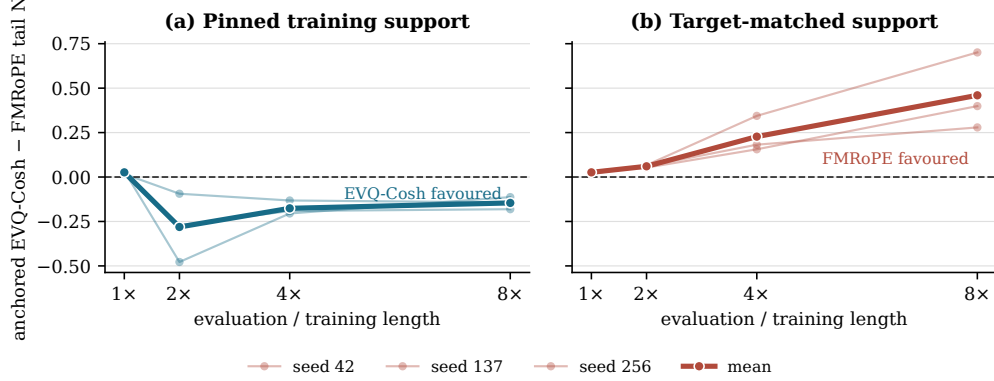


Figure 6: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

Under fixed, anchored EVQ-COSH is favoured by 3/3 seeds at every OOD length; under target-matched, FMRoPE is favoured by 3/3 seeds at every OOD length. The fixed-range sign agrees across all seeds and OOD lengths; the 1K/2K magnitudes are the most stable across seeds.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. Under the tested target-retargeted policy, the ordering of the two tested policies reverses. The two columns therefore establish that the fixed-support result is conditional on the selected range; they do not identify a support-by-allocation interaction law. The *fixed* column isolates allocation when one table serves multiple lengths (§2.1).

C.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMRoPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

Table 10: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	FMRoPE-derived table	Anchored-EVQ-COSH-derived table
FMRoPE-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. This is a two-seed mechanism replication of weight-table co-adaptation, not a replacement for the three-seed exact-range effect and not evidence that either derived table is globally optimal.

C.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at $0.75\times$, $1.00\times$ (the formula point), $1.25\times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75\times$ 6.202858 (494.16); EVQ-COSH $1.00\times$ 6.202094 (493.78); EVQ-COSH $1.25\times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5\times$ arm is better; at formula $\tau=8$ the $0.75\times$ arm is. The improving direction points inward at both tested extremes; these discrete arms do not locate continuous basin boundaries or yield a replacement closed-form rule.

Table 11: **Exact-range factorial** (50.9M; $2 \text{ bases} \times 2 \text{ training lengths} \times 3 \text{ head dimensions} = 12$ configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times / 4 \times / 8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	−0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

Scope. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated WikiText-2 raw stream (Merity et al., 2017), a different corpus from the 151.9M control and from the mature-scale runs. The factorial establishes cross-configuration *direction*, while the 151.9M control provides the effect size within its configuration. Scale is evaluated in §4.4.

D LEARNED-FREQUENCY COMPARATOR

Table 12: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8\text{K}$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). Allocation-shape attribution uses the fixed, zero-parameter analytic schedules of §2.1; this separately reported arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026); DAPE belongs to the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 12 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at 10 $\times$ and 100 $\times$ the base rate, and the stronger 100 $\times$ setting is the 455.3 row. It answers the learned-table tuning question. Allocation-shape attribution uses the fixed analytic schedules under an identical zero-parameter operator in §2.1.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). In this protocol, the learned table improves over Geo but remains weaker than EVQ-COSH; the comparison does not identify whether the gap comes from the final table, optimization dynamics, or the positional-parameter learning rate.

E MATURE-SCALE PROTOCOLS

E.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the endpoint grid $u_k=k/K$ with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (OLMo Team, 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-

Table 13: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K *niah_single_1*, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive end-points are kept separate in the discussion below and in Table 13.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled Q/K *metrics.json* records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 13 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals condition on the single trained pair and quantify evaluation-row uncertainty, not training-seed variability.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/ +0.0381/ -0.0437/ -0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe natural-text LM comparison; capability is evaluated by the separate autoregressive protocols above.

The identity is scientific rather than bitwise trainer pairing. The EVQ-COSH branch uses the reviewed Hugging Face single-GPU loop, whereas the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context,

1,000 optimizer steps, and 2,097,152,000 counted input tokens. Different trainer stacks remain a residual implementation confound, so this is scale persistence from one same-initialisation/same-recipe pair, not a bitwise paired trajectory or a multi-seed causal estimate.

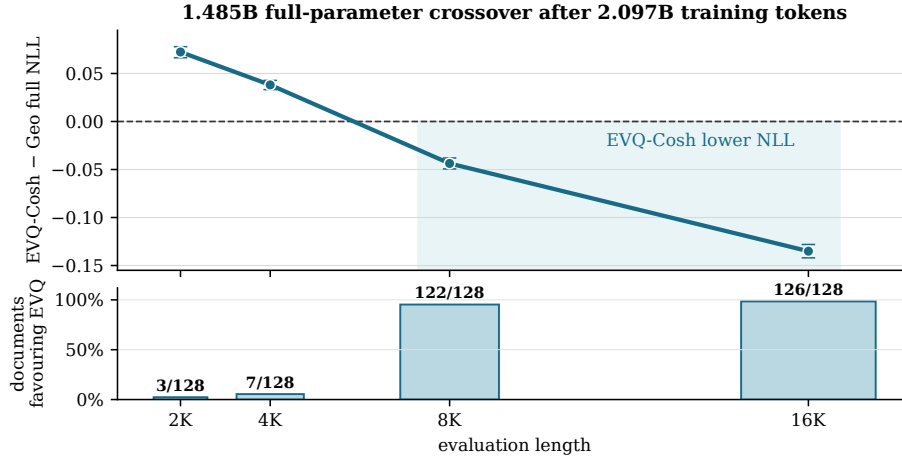


Figure 7: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

E.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Fig. 2 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The two reference rows do not: Native retains the pretrained table, while official Transformers YaRN applies its frequency-dependent transport. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $m_k = (1 - \tilde{U}_k)^2$ in Eq. (8), so less unique pairs move farther.

Table 14: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Fig. 2. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalue, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024)

uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are [54.88, 64.80] points for OLMo and [0.25, 17.50] points for Qwen; the paired derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 15 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 15: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The complete OLMo deployment policy is consequently a separate bundled intervention: exact Native is selected when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Table 16: Compact zero-training deployment endpoints on the frozen OLMo checkpoint. These are complete-policy comparisons, not pure- z effects.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$, so the result is not universal task dominance. The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Figure 2 remains the allocation-only contrast.

E.3 LLAMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table. The training objectives, adapters, and endpoints remain separate below.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. The contrast is therefore the end-to-end effect of assigning different frequency substrates under a

matched adaptation pipeline; it does not isolate pure interior shape because the Native endpoint grid and midpoint-Cosh grid also differ in finite-grid embedding. Under this protocol, EVQ-COSH-minus-Native NLL is $+0.390/-1.510/-2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry; it changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 13 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, establishing task-family-adapted length transfer. The natural-text and routing rows evaluate probability and causal source use.

F SUPPORTING RESULTS

F.1 MULTI-HEAD LATENT ATTENTION

We test the finite-budget hypothesis in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified empirical $\tau=1.414$ setting. Because the compressed rotary subspace does not instantiate the full-RoPE $d_{\text{rot}}=d_{\text{head}}$ convention in Eq. (37), this experiment tests frequency allocation at fixed architecture rather than the operating rule.

The tracked aggregate is backed by the exact recovered three-seed evaluator JSON and retains every seedwise row. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree. Original checkpoint and training-data artifacts are not packaged, so this row establishes the reported system ordering, not checkpoint-level reproduction or a theorem fixing τ from d_{rope} .

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 17: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond. These rows establish the tested ordering for this

1620 architecture and protocol; the multiplier overlay is the run-specific MLA wavelength-blend operator,
1621 not the paper's pure- z control or an exact reproduction of another scaling method.
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